



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI
INSTRUCTION DIVISION

FIRST SEMESTER 2026-2027

Course Handout (Part - II)

Date: 1st Aug., 2026

In addition to part-I (General handout for all courses appended to the time table) this portion gives further specific details regarding the course.

This course is offered as an elective for the first degree and higher degree students, as indicated in the pool of discipline electives under respective programmes. It can also be taken by other discipline students as open elective, as applicable and eligibility.

Course No. : BITS F467 / BITS U467
Course Title : BIOETHICS AND BIOSAFETY
Instructor-in-Charge : R. MAHESH

1. Scope and Objective of the Course:

This course is designed to create an awareness on the basic concepts of biosafety and bioethics that are essential for various branches of science which has had an great impact on quality of life and regulations governing the experimentations on transgenic organisms and plants, prenatal diagnosis and related social issues, handling and disposal of hazardous materials, environmental pollution concerns and ethics in need based experimentation on animals and plants, would be addressed. Emphasis will also be made on health care related issues.

2. Text Books (TB):

S. Ignachimuthu, S.J., **Bioethics**, Narosa Book Distributors Pvt. Ltd., Delhi, 2009 **or latest edition**
(As text material may not cover latest details, for details on recent developments related to the topic, students are advised to consult the Instructor-In-Charge for guidance to obtain information or obtain through search activities on their own).

3. Reference Books (RB):

- a) M.K. Sateesh, **Bioethics and Biosafety**, I. K. International, Pub. House Pvt.Ltd., New Delhi, 2008
- b) Ben Mephram, **Bioethics**, II Edn., Oxford University Press, Hampshire, Great Britain, 2008





4. Course Plan:

Learning objectives and Topics	Lectures*	Book(s) for Reference
Introduction to the need and issues governing biosafety	1-3	TB:1; RB-2:1
Legal, ethical and social implications including issues related to human reproduction, gene manipulation, genetic testing	4-6	TB:2,3 RB-2:2,5,8
Guidelines for research in transgenic organisms and plants such as cloning, stem cell research, gene therapy, etc.	7-11	TB:2,3; RB-2:3
Ethics in stem cell research	12-14	TB:3
Animal cloning and organ transplants	15-17	TB:3,4; RB-2: 26
Biosafety guidelines, regulations, public participation	18-20	TB:4.2, 5.1,5.3; RB-2:12,27
Socio-economic impacts of biotechnological experiments	21-22	RB-2:8
Environmental pollution- hazards and control (including Disinfection, Sterilization, Sanitation methods)	23-30	TB:5.3; RB-2:9,10
CPSEA guidelines	31-34	TB:4.3, 5.1-5.2; RB-2:13-15
GLP and GMP guidelines	35-39	RB-2:11
Introduction to IPR and Patent processing	self study@	RB-2: 16-17
@Buffer lecture session for discussion on select topics		As required
* Sequence of lectures on above topics could be changed based on student understanding and prior knowledge		

* The lectures may be slightly deviate (including sequence of coverage), from aforementioned plan, based on students' background and interest in the topic, which may perhaps include special lectures*** and discussions, that would be planned and schedule notified, accordingly.

Assignment / Practical / Field / Case Studies

The Assignment / Practical will be given conducted in on either some or all of the below mentioned topics. Details will be intimated through a separate notification or in class. NO MAKE-UPS WOULD BE ALLOWED FOR Quizzes, SUBMISSION OF ASSIGNMENTS / PRACTICAL REPORTS, WHICH MIGHT INCLUDE FIELD, data collection and analysis, Lab. study.





5. Evaluation :

	Evaluative Component	Weightage (in %)	Duration	Date and Time	Comments
1	Mid Semester Test	30	90 mins	As announced by AUGSD/AGSRD	Closed book
2	Continuous Assessment of classroom participation and discussions and resulting surprise quiz(zes)#	10	continuous	As announced in class by I/C	Open Book & or Closed Book
3	Assignments, seminars, practical reports, case studies, field study, data collection and analysis, Lab. data study. (practicum experience)	25	continuous	As announced in class by I/C	Open Book & or Closed Book
3	Comprehensive Examination	35	180 mins. Minimum	As announced by AUGSD/AGSRD	Open Book & or Closed Book

*Case studies, interpretation of data and then analysis, will form a part of all evaluation components. Assignments(s) may include seminar presentation and viva. Assignment(s)/Quizzes may be practical / theory oriented, for which a proper documentation and submission of records and reports in a standard format should be submitted as per deadline(s) that would be announced. It may also include a viva and or seminar presentation(s).

* Assignment(s) may be field work / practical / theory oriented, for which one copy of **hand-written/ type-written report, as will be announced in class, in a standard format** should be submitted as per deadline(s) that would be announced, therein. It may also include a viva and or a seminar presentation. However all assignments/reports would be completed by **last week of Nov., 2025 (date will be intimated)**. It is necessary that all students **stick to time schedule and DO NOT postpone submission** of assignments/reports. **[No report will be received / considered for evaluation sent through mail unless and otherwise specified]**. This will prevent extra load during last two weeks of class work. No make-up will be given for time defined assignments, field studies and **NO MAILS will be sent**. It is the responsibility of the student to be aware of class-room discussions. . Anyone not present in class during such evaluatives will be considered as ABSENT and no marks will be given.





Quiz(zes) may/will be conducted as a part of evaluation component, at random, during contact hours including lecture, tutorial hours, with/without prior intimation, and sometimes outside class contact hours (pre-announced), and hence it is expected that the students come prepared to every class on topics covered in earlier contact hours. Students are also requested to refresh their **knowledge in basic subjects, as relevant.** No make-ups will be possible for the same.

\$ It is strongly advised that all students prepare their own class notes and relevant information from text, reference material, as in handouts would only be allowed for consultation during assessments of open book/notes. Photocopies of any material, written or printed will not be permitted. Stapled sheets, loose sheets of information written or printed, photocopies of slides used for discussion in class, will not be allowed for evaluatives, unless otherwise specified. Someone else's notes will also not be permitted. No other text/reference material (including edition mentioned in handout) will be permitted for reference during open book assessments.

*Slides used during class hours provide key information for which additional supportive information is expected to be collected from sources aforementioned. **These slides will not be shared through mail and hence students are requested/advised to make /prepare their own notes during class hour/outside class hour.** Recent developments in the area/topic will be discussed in class based on their significance to the course and hence some information being latest developments in the field discussed in classroom may differ from the information in text, reference material and hence students are expected to take note of such key discussions during contact hours. Such discussions held in class will be considered as primary source of information in assessments. Also note that all classes will be held (online/in person) as announced by AUGRD/AUGSD, from time to time and students are requested to note the same.

* Attendance is equally important which may be directly / indirectly related to grading.

* All evaluation components are equally important, irrespective of weightage. Hence, students failing to attend or absenting themselves in one or many of the evaluation components or classes, may become ineligible for obtaining a valid grade at the end of the semester. Therefore, students are strongly advised to keep away from absenting themselves from all aforementioned contact sessions. Clearing the course would require adequate performance in written quiz(zes)** , tests** , examinations** , assignments, separately. It is the responsibility of the student to be aware of all discussions that happen during lecture/tutorial/ other contact sessions.





For all evaluation components, information given in class lectures, tutorials, text books and reference books, in the same order will be considered, as correct. All evaluation components are equally important, irrespective of weightage. Hence, students failing to attend or absenting themselves in one or many of the evaluation components, including participation in classroom discussion, active participation in laboratory sessions may become ineligible for obtaining a valid grade, at the end of the semester. Attendance in lectures, tutorials, other evaluatives like quiz(zes) and practicum experience, are all equally important as they are all integral components of learning, **irrespective of weightage** and may be taken into consideration, during grading.

Students are strongly advised to keep away from absenting themselves from all aforementioned contact sessions. Clearing the course would require adequate performance in written quizzes**, tests**, examinations**, and in practicum experience components, separately (i.e. procuring low marks in evaluation components, aforementioned**, other than practicum experience, would not suffice, to clear the course). All announcements regarding the course will be mostly discussed / shared during class/lab. contact hours and no mail may be sent to individual student. It is the responsibility of the student to be aware of all discussions that happen during lecture/tutorial/lab. contact sessions.

Any other adaptive changes in the handout, will be announced in class, if any.

6. Reading Assignments :

Students are advised to read, collect additional information on the above mentioned topics as per given schedule. In addition, awareness w.r.t. latest developments in the area would be an added advantage.

* As this is an elective course, more stress will be laid upon **latest information available from different primary and secondary sources** such as latest edition of books, journals, articles published and research findings, reported in standard literature, independent studies, newsletters, etc. The chapters mentioned in the text-books would give basic information related to topics whereas the class room information would give latest developments and their applications in the field. Hence, only reference to text / reference material would not suffice and students are advised to take a note of the same.





7. Chamber consultation hours : (3170-S) ; Time: To be announced in class, based on mutual availability.
8. Notices : Notices concerning the course will be displayed on the **Pharmacy Department N.B. only** and shared through e-mail, only as necessary. Not all notices will be sent by mail.
9. Mid.-Sem. Grading : Mid.-Sem. Grading will be displayed whenever about 30-40 % of evaluation components are completed.
10. Make-Ups :

Make-Ups are not given as a routine. It is solely dependent upon the GENUINENESS OF THE CIRCUMSTANCES under which a student fails to appear in a scheduled evaluation component. In such circumstances, prior permission should be obtained from the Instructor-in- Charge. In no case the make-up letter be slipped inside the chamber of the Instructor-In-Charge or intimation sent after the evaluative (applicable only for mid-sem. test and comprehensive examination). The decision of the Instructor- in-Charge in these aspects will be final. No messaging or mails will be entertained as students being residential on campus, can met the I/C of the course, in terms of any difficulty.

Instructor - in -Charge
BITS F467

